

Shubhangi S. R. Garnaik

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My research focuses on human-centered artificial intelligence, particularly multimodal interaction, augmented reality, speech systems, and trustworthy AI. I develop deployable systems that balance performance, latency, robustness, and usability in real-world environments.

Research areas: multimodal AI, conversational agents, augmented reality, computer vision, trustworthy and deployable AI

Research Experience

2024–Present	Graduate Researcher, AI2S Lab Stony Brook University - Develop human-centered multimodal systems spanning conversational AI, augmented reality, and audio-visual interaction. - Build and evaluate deployable prototypes with an emphasis on grounded responses, low latency, reliability, and user trust.
Jul.–Dec. 2025	Research Intern, MEIC Lab Stony Brook University - Implemented and containerized a 3D Gaussian Splatting workflow covering preprocessing, training, postprocessing, and rendering. - Designed edge-case tests for backend APIs and documented failure modes to improve research-software reproducibility.
2025	Research Assistant, School–Company Collaboration Taeyang Metal Industrial Co., Ltd. - Developed computer-vision workflows for real-time press and forging monitoring from analog gauges and digital graph displays. - Analyzed production recordings and image-quality constraints to support anomaly detection and equipment-failure prediction.

Professional Experience

Mar.–Oct. 2024	Data Scientist Thanks Carbon, Republic of Korea - Developed predictive models using weather and rice-yield data to improve agricultural efficiency and support reduced carbon emissions. - Applied deep-learning methods to regional predictive analytics for alternate wetting and drying practices.
Jun. 2019–Sep. 2021	Software Test Engineer / Associate Technical Sales Engineer Zensoft / Qualitia, India Automated client software testing and created structured validation workflows to reduce testing time and improve software quality.

Education

2024–Present	Ph.D. in Computer Science Stony Brook University Advisor: Professor Jihoon Ryoo
2022–2023	M.S. in Computer Science Stony Brook University GPA: 3.92/4.00; supported through university and advisor scholarships
2015–2019	Bachelor of Engineering in Computer Engineering Savitribai Phule Pune University GPA: 3.33/4.00

Publications

2026	When Machines Speak Like Local Peers: Improving Conversational Experiences with Accent-Adaptive Voice Agents Shubhangi S. R. Garnaik, JiHyun Jeong, and Jihoon Ryoo
Interspeech, accepted	paper • project page
2025	ArtSpeak: An Interactive AR Application for Lifelike Speaking with Art Portraits Shubhangi S. R. Garnaik, Aruna Balasubramanian, Niranjana Balasubramanian, and Jihoon Ryoo
IEEE ISMAR	IEEE Xplore • paper • project page
2025	CLOUD-CODEC: A New Way of Storing Traffic Camera Footage at Scale Hoyoung Kim, Azimbek Khudoyberdiev, Shubhangi S. R. Garnaik, Arani Bhattacharya, and Jihoon Ryoo
ACM TOMM	DOI: 10.1145/3744649
2023	A Study of the Man and Unmanned Teaming System for Improving Efficiency in a Fulfillment Center Azimbek Khudoyberdiev, Shubhangi S. R. Garnaik, Ha Young Kim, and Jihoon Ryoo
IEEE Access	DOI: 10.1109/ACCESS.2023.3341396
2023	MINSU: Precision Quantity Counter with DNN-based Volume Estimation Shubhangi S. R. Garnaik, DongYeop Lee, Youngho Kim, and Jihoon Ryoo
IEEE ICTC	DOI: 10.1109/ICTC58733.2023.10392667
2022	SQR: Secure QR Transaction with Randomized Rotation Shubhangi S. R. Garnaik, Youngho Kim, and Jihoon Ryoo
IEEE ICTC	DOI: 10.1109/ICTC55196.2022.9952603

Posters and Workshop Papers

2023	Enhancing Liver Disease Prediction Using Machine Learning Models Comparative study of oversampled and synthetic-data approaches; presented at KICS.
2023	Exploring Sentiment Analysis of the Silicon Valley Bank Collapse Deep-learning approaches applied to Google News headlines; presented at KICS.

Teaching Experience

2025–2026	Teaching Assistant Stony Brook University CSE 113: Foundations of Computer Science; CSE 310: Computer Networks; CSE 373: Analysis of Algorithms
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Honors and Awards

May 2026	Winner, Decompiler HACKATHON THE X Recognized for a winning project at the 2026 hackathon.
2019	Idea Finalist, Flipkart Hackathon Selected as a finalist for an industry innovation challenge.
2019	Conference Scholar, Distributed Systems Conference Awarded conference scholarship support in Pune, India.

Technical Skills

Programming

Python, Java, C#, HTML/CSS

AI/ML and Vision

PyTorch, TensorFlow, OpenCV, deep learning, computer vision, video analytics, 3D Gaussian Splatting, snnTorch, Lava

Systems and Tools

Git/GitHub, Linux, Docker, Jupyter, Google Colab, Unity, Android Studio, VS Code, Eclipse

Testing and Communication

Backend API testing, software test automation, model documentation, reproducible workflows, technical presentations

Service and Leadership

Orientation Leader

Supported incoming students during their transition to university life.

Competition Organizer

Organized an inter-college coding competition in Pune to promote computational problem solving.